# 22PH202 ELECTROMAGNETISM AND MODERN PHYSICS
# UNIT III – LM-3.6 LASER

## 1 INTRODUCTION- LASER

The word LASER is acronym for **LIGHT AMPLIFICATION BY STIMULATED EMISSION OF RADIATION**. Laser source produces coherent, monochromatic, least divergent, unidirectional and high intense beam. Einstein gave the theoretical description of stimulated emission in 1917. In 1954 G.H. Towne developed microwave amplifier MASER using Einstein's theory and put forward to light and first Laser was developed.

## 2. CHARACTERISTICS

### 2.1 DIRECTIONALITY

Ordinary light spreads in all directions and its angular spread is 1m/m. But, it is found that laser is **highly directional** and its angular spread its 1mm/ meter. For example the laser beam can be focused to very long distance with a few divergence (or) angular spread.

Divergence (or) Angular spread is given by $(\phi) = (r_2 - r_1) / (d_2 - d_1)$ degrees

Where $d_1$ & $d_2$ are any two distances from the laser source emitted and $r_1$, $r_2$ are the radii of thebeam spots at a distance $d_1$ & $d_2$ respectively

### 2.2 INTENSITY

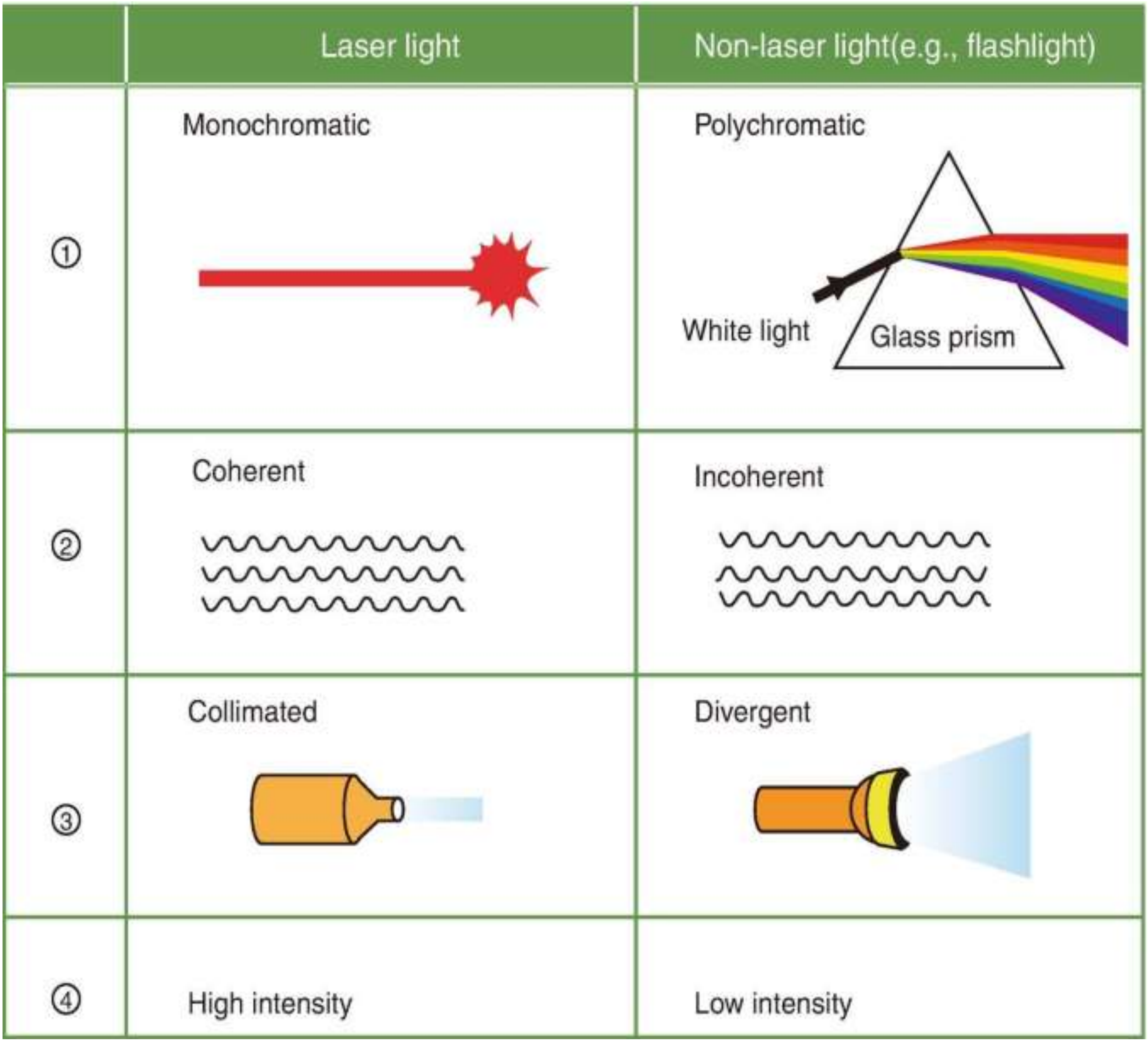
Since an ordinary light spreads in all directions, the intensity reaching the target is very less. But in the case of laser, due to high directionality the intensity of laser beam reaching the target is of **high intense beam.** For example, 1mW power of He – Ne laser appears to be brighter than the sunlight

### 2.3MONOCHROMATIC

Laser beam is **highly monochromatic** (i.e.,) the wavelength is single, whereas in ordinary light like mercury vapour lamp, many wavelengths are emitted.

### 2.4 COHERENCE

In lasers the wave trains of same frequency are in phase (i.e) the radiation given out is in mutual agreement not only in phase but also in the direction of emission and polarization. Thus it is a **Coherent beam.**



| | Laser light | Non-laser light(e.g., flashlight) |
|---|---|---|
| ① | Monochromatic | Polychromatic (White light, Glass prism) |
| ② | Coherent | Incoherent |
| ③ | Collimated | Divergent |
| ④ | High intensity | Low intensity |

# 3 THEORY OF LASER

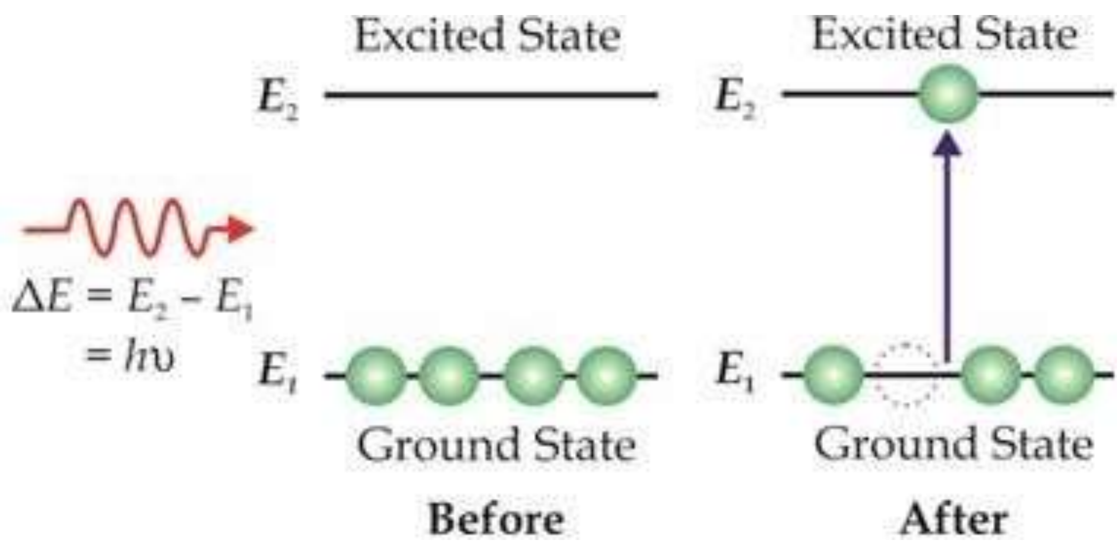
## 3.1 INTERACTION OF LIGHT RADIATION WITH MATERIALS

Consider an assembly of atoms in a material which is exposed to light radiation (a stream of photons with energy *hv*)

In general, three different processes occur when light radiation interacts with material. They are: **(1) Stimulated absorption (2) Spontaneous emission (3) Stimulated emission.**

## 3.2STIMULATED ABSORPTION

At atom in ground state with energy $E_1$ absorbs an incident photon of energy *hv* and is excited to higher energy state with energy $E_2$. This process is called stimulated (or) induced absorption. It occurs only when the incident photon energy *hv* is equal to the energy difference between excited state and ground state ($E_2 - E_1$).

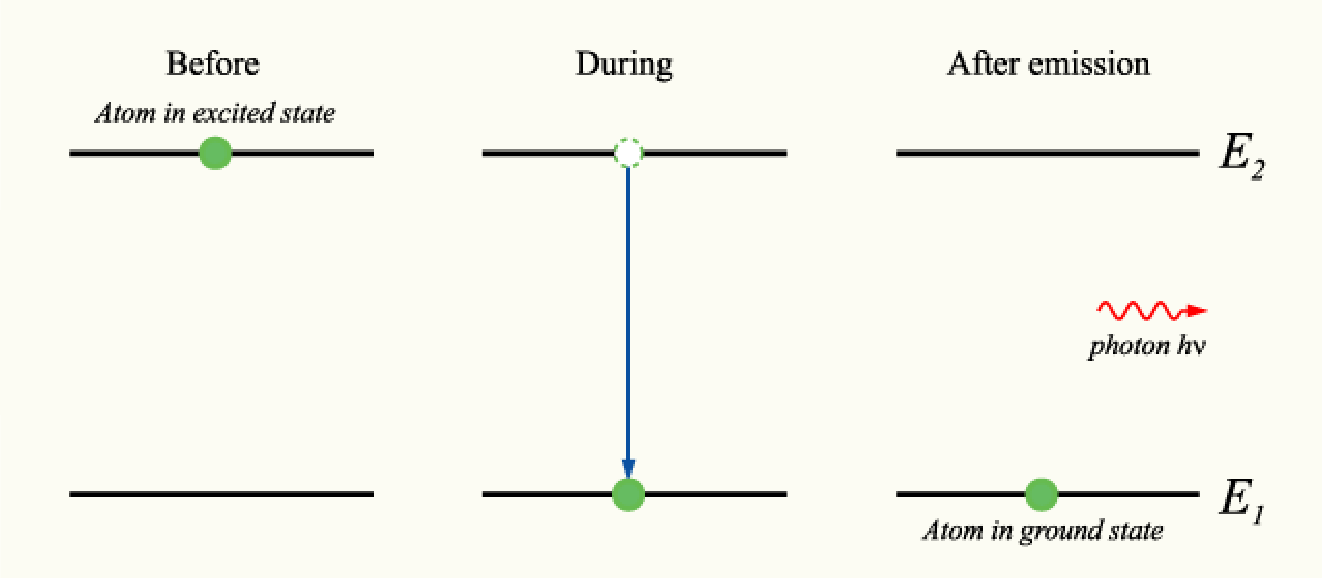



**Stimulated absorption**

For each such a transition, a certain amount of energy *hv* is absorbed from the incident light beam. The excited atoms do not stay in the higher energy state for a longer time. It is the tendency of atoms in excited state to come to the lower energy state. Thus, the atoms in excited state quickly return to ground state by emitting a photon of energy *hv*. The emission of photons takes place in two ways : (a) Spontaneous emission (b) Stimulated emission

## 3.3 SPONTANEOUS EMISSION

The atom in the excited state $E_2$ (higher energy state) returns to ground state $E_1$ (lower energy state) by emitting a photon of energy *hv* (□E = $E_2 - E_1$) without the influence of any external agency. Such emission of light radiation which is not triggered by any external influence is called spontaneous emission. It is a random and also uncontrollable process.

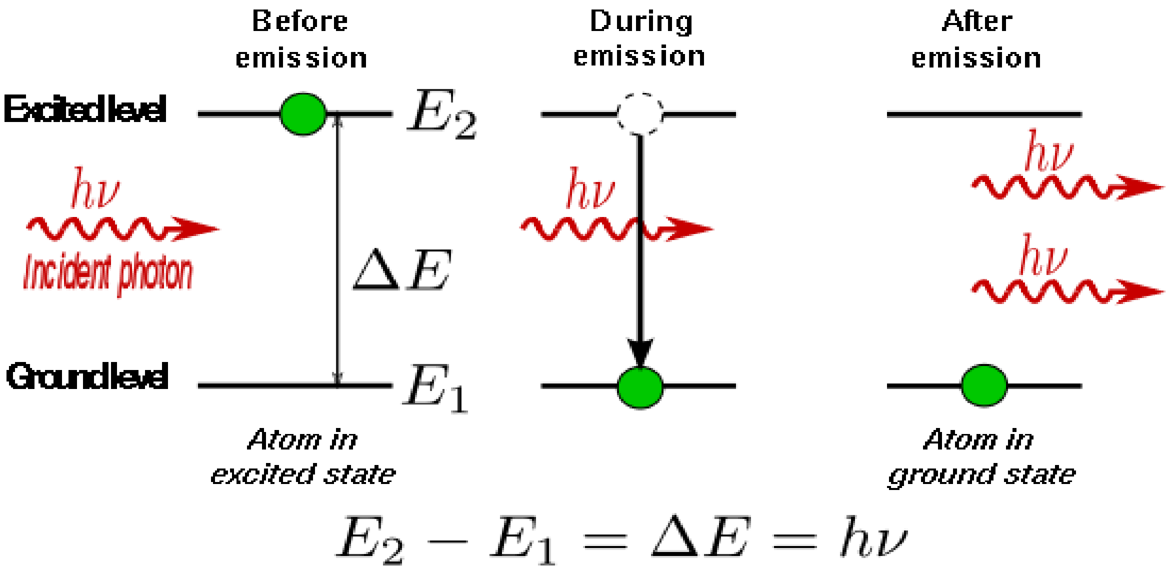



**Spontaneous emission.**

## 3.4 STIMULATED EMISSION

Einstein suggested that there must be another mechanism by which an atom in excited state can return to ground state. He found that there is an interaction between the atom in excited state and a photon. During this interaction, the photon triggers the excited atom to make transition to ground state E1. This transition produced a second photon which is similar to triggering photon with respect to frequency, phase and propagation direction.

Such kind of forced emission of photons by the incident photons is called stimulated emission. It is also known as induced emission. It plays a key factor for the working of a laser.

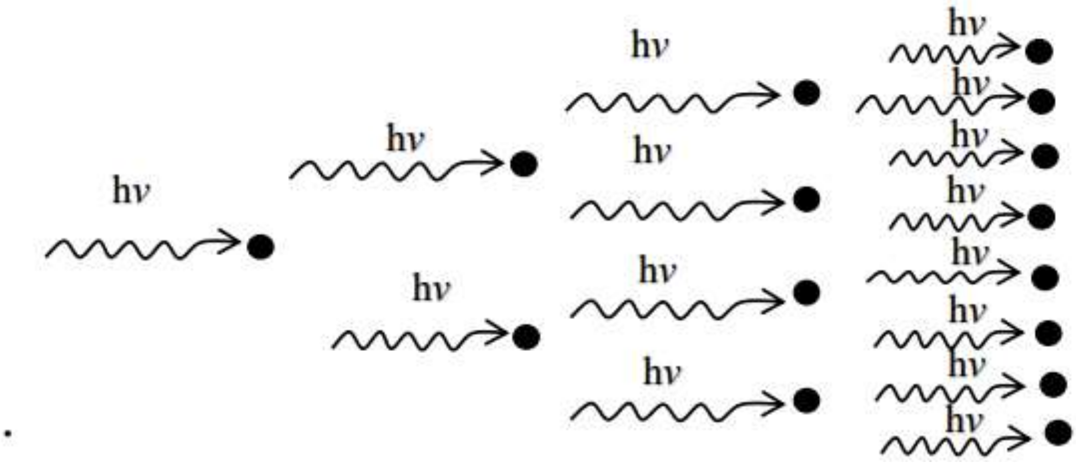



**Stimulated emission**

# 4 CONCEPT OF LASER

The photon emitted during stimulated emission has the same energy, phase, frequency and direction as that of the incident photon. Thus, we have two coherent photons in the above case. These photons now incident on two other atoms in the state $E_2$. This will result in induced emission of two more photons. Now there are four coherent photons of same energy. These four photons may induce further transitions with four other atoms in the energy state $E_2$. This gives rise to stimulated emission of eight coherent photons of same energy.

f the process continues in a chain, we will ultimately be able to increase the intensity of coherent radiation enormously. Stimulated emission is multiplied through a chain reaction. The multiplication of photons through stimulated emission leads to coherent, powerful, monochromatic, collimated beam of light. This light is known as LASER. Laser requires stimulated emission exclusively. This can be achieved by population inversion




**concept of laser**

## 4.1 POPULATION INVERSION

Population inversion creates a situation in which the number of atoms in higher energy state is more than that in lower energy state. Usually at thermal equilibrium, the number of the atoms in the higher energy state $N_2$ is much smaller than the population of atoms at lower energy state $N_1$. i.e., $N_1 > N_2$.

The phenomenon of making the number of the atoms in the higher energy state greater than that of the lower energy state is called population inversion.

**Condition**

There must be two energy levels

There must be a source to supply the energy

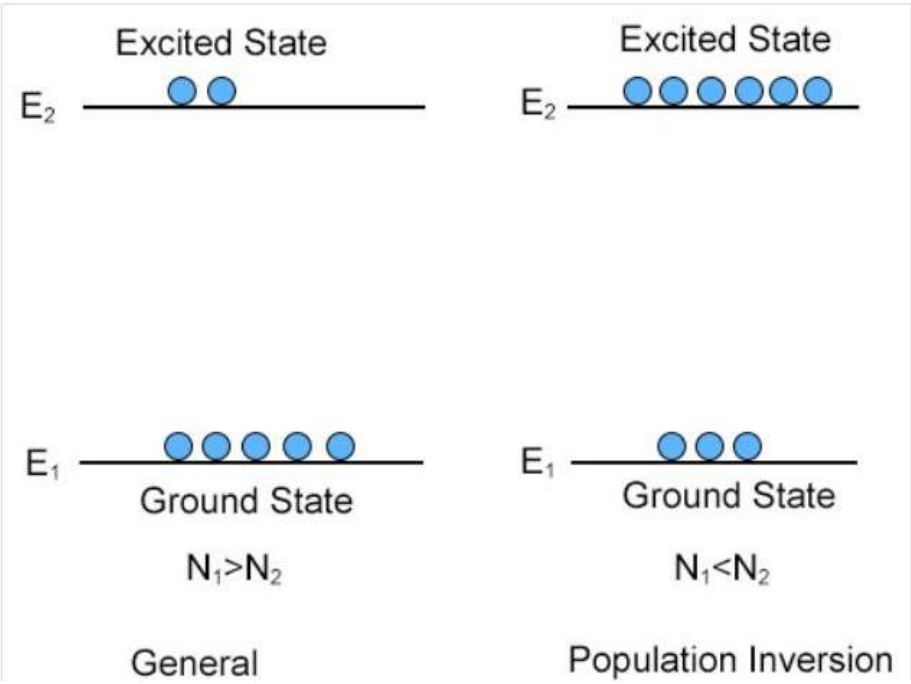
The atoms should be continuously raise to excited state

**Active medium**

A medium in which population inversion can be achieved is known as active medium.

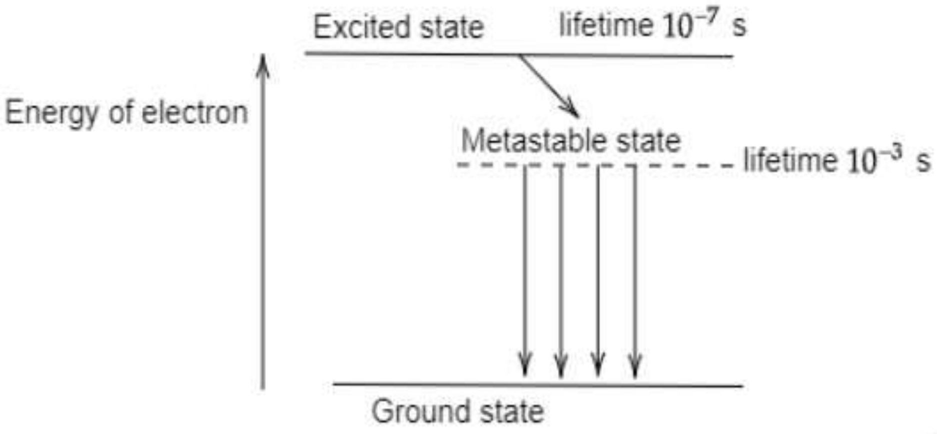
**Pumping Action**

The process to achieve the population inversion in the medium is called pumping action.




## 4.1.2 METASTABLE STATES:

An atom in the exited state has very short life time which is of the order of $10^{-8}$ *sec*. Therefore even if continuous energy is given to the atoms in ground state to transfer them to exited state they immediately comes back to the ground state. Thus population inversion cannot be achieved. To achieve population inversion, we




ENERGY LEVEL DIAGRAM

must have energy states which has a longer lifetime. The life time of meta stable state is $10^{-3}$ to $10^{-6}$ which is $10^{3}$ to $10^{6}$ time of exited states thus allows accumulation of large number of excited atoms and result in population inversion.

### 4.1.1 METHODS

(1) Optical pumping (excitation by photons)

(2) Electrical discharge method (excitation by electrons)

(3) Direct conversion

(4) Inelastic collision between atoms

#### (1) OPTICAL PUMPING

When the atoms are exposed to light radiations of energy *hν*, atoms in the lower energy state absorbs these radiations and they go to the excited state. This method is called Optical pumping.
Eg: Nd - YAG Laser, Ruby laser

#### (2) ELECTRICAL DISCHARGE METHOD

In this method, the electrons are produced in an electrical discharge tube. These electrons are produced in an electrical discharge tube. These electrons are accelerated to high velocities by a strong electrical field. These accelerated electrons collide with the gas atom.

By this process, energy from the electrons is transferred to gas atoms. Some atoms gain energy and they go to excited state. This results in population inversion. This method is called electrical discharge method.

$A + e^{*} \rightarrow A^{*} + e$

A – gas atom in ground state

$e^{*}$ - Electron with high kinetic energy$A^{*}$
- same gas atom in excited state
E – Electron with lesser kinetic energy

**(3) DIRECT CONVERSION**

In this method, due to electrical energy applied in direct band gap semiconductor like GaAs, recombination process, the electrical energy is directly converted into light energy.

Electrical → Light energy

**(4) INELASTIC ATOM – ATOM COLLISION**

In this method, a combination of two gases is used say A & B. The excited states of A & B nearly coincides in energy. During electrical discharge atoms of gas A are excited to metastable state A* due to collision with electrons.

$\mathbf{A + e^{*} \rightarrow A^{*} + e}$

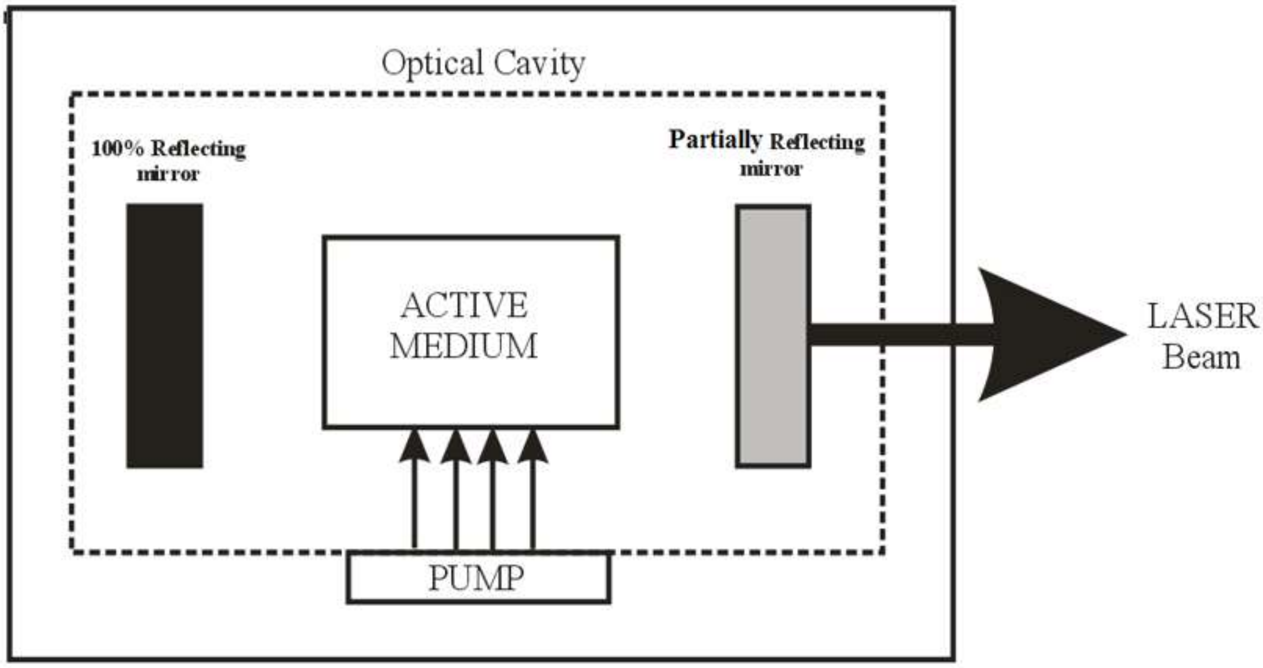
Now, A* atoms at metastable state collide with B atoms in the lower state. Due to this inelastic collision B atoms gain energy and they are excited to a higher state (B*). Hence , A atoms lose energy and return to lower state

$\mathbf{A^{*} + B \rightarrow A + B^{*}}$

This result is population inversion in the energy states of B.

## 4.1.2 BASIC COMPONENTS OF LASER SYSTEM

A laser system consists of three important components. They are




**Basic components of laser system**

Active medium (or) active material

(a) Pumping source

(b) Optical resonator

**(a) ACTIVE MEDIUM (OR) ACTIVE MATERIAL**

It is a medium in which atomic transitions takes place to produce laser action. The active medium may be solid, liquid, gas, dye or semiconductor.

**(b) PUMPING SOURCE**

It is a system used to produce population inversion in the active medium.

**(c) OPTICAL RESONATOR**

An optical resonator consists of a pair of reflecting surfaces in which one is fully reflecting ($R_1$) and the other is partially reflecting ($R_2$). The active medium is placed in between these two reflecting surfaces

The photons generated due to stimulated emission are bounced back and forth between these two reflecting surfaces. This induces more and more stimulated transition leading to laser action.

# 5 TYPES OF LASER

Based on the type of active medium, laser systems are broadly classified as follows:

| Sl.No | Type of laser | Examples |
|---|---|---|
| 01. | Solid state laser | Ruby laser, Nd-YAG laser |
| 02. | Gas laser | He – Ne Laser,$Co_2$ laser, Ar- ion laser |
| 03. | Liquid laser | $SeOCl_2$ laser. |
| 04. | Dye Laser | Rhodamine 6G laser |
| 05. | Semiconductor laser | GaAs Laser, GaAsP laser |

# 6. APPLICATION OF LASER

1) The laser beam is used to vaporize unwanted materials during the manufacturing of electronic circuitson semiconductors chips.
2) Laser is used to detect and destroy the enemy missiles during war.
3) Metallic rod can be melted and joined by means of laser beam.
4) Low price semiconductor lasers are used in CD players, laser printers.
5) High power lasers are used to leasing thermo nuclear reactions which would become the ultimateexhaust little power source for human civilization.
6) Laser is also being employed for separating the various isotopes of an element.
7) Laser beam are also been used to the internal confinement of plasma.